V.L Memorial Public School

475 Vijay Nagar

CHAPTER-4-QUESTIONS-CLASS6TH-SCIENCE

Class 06 - Science (Curiosity)

Section A

1. A magnet attracts **[1]**

   a) nickel, iron, cobalt
   b) book, pen, pencil
   c) iron, nickel, plastic
   d) bench, chair, cobalt

2. Which of these materials can be used to make a temporary magnet? **[1]**

   a) a piece of plastic
   b) a piece of wood
   c) a piece of paper
   d) a piece of iron

3. What happens when the North pole of one magnet approaches the North pole of another? **[1]**

   a) They attract each other
   b) They stop being magnets
   c) They repel each other
   d) They lose their poles

4. Why was the matsya-yantra useful for navigation at sea? **[1]**

   a) It measured the speed of ships
   b) It measured the depth of seawater
   c) It used a magnet to indicate direction
   d) It stored water during sea journeys

5. What happens when a magnet is heated? **[1]**

   a) It causes nearby objects to attract stronger.
   b) It becomes stronger.
   c) Its magnetic property is lost or reduced.
   d) It breaks into smaller magnets.

6. What will happen when the two magnets shown below are pushed towards each other? **[1]**



   a) Nothing will happen
   b) They will attract each other
   c) They will repel each other
   d) They will heat up

7. Which of the following is a combination of non-magnetic materials? **[1]**

   a) Plastic, wood, iron and paper
   b) Plastic, leather, clothes and paper
   c) Leather, iron, nickel and clothes
   d) Iron, nickel and cobalt

8. Why would a magnetic compass be useful during an overcast night? **[1]**

   a) It helps measure rainfall
   b) It helps predict storms

c) It helps find directions
d) It helps store spices

9. Why do most iron filings gather near the ends of a bar magnet? **[1]**

a) The filings become magnetic there
b) The magnet has poles there
c) The middle is strongest there
d) The poles are strongest there

10. Magnets can be protected against self-demagnetisation by keeping the two magnets **[1]**

a) over each other lengthwise.
b) with their similar poles on the same side and soft iron keeper in between
c) opposite poles with keepers and wood piece
d) at angles to each other

11. The poles of a magnet are located at its **[1]**

a) middle point
b) ends
c) lower surface
d) upper surface

12. What indicates the presence of two poles in a magnet? **[1]**

a) It only attracts other magnets
b) A magnet always has both a North and a South pole
c) It can be divided into single poles
d) The existence of only a North pole

13. Which statement is true about non-magnetic materials? **[1]**

a) They are usually metals.
b) They include iron, nickel, and cobalt.
c) They are attracted towards magnets.
d) They do not respond to magnets.

14. Why does a freely suspended magnet settle in the north-south direction? **[1]**

a) The magnet rests on a dial
b) Earth behaves like a giant magnet
c) The magnet has a circular shape
d) The magnet contains an iron needle

15. How should magnets be stored to retain their strength? **[1]**

a) In pairs with unlike poles on the same side
b) Tethered by strings
c) Lying freely as single pieces
d) When alone, wrapped in cloth

16. Which of the following actions should be avoided to maintain a magnet's effectiveness? **[1]**

a) Covering it with a piece of wood
b) Keeping it near mobile phones
c) Dropping it frequently
d) Placing it near another magnet with unlike poles facing each other

17. What color is the South pole on certain magnets? **[1]**

a) Red
b) Green
c) Blue
d) Yellow

18. **Assertion (A):** Bar magnets are kept in pairs with their unlike poles on the same side. **[1]**
Reason (R): Magnets become weak if they are not stored properly.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

19. **Assertion (A):** Magnetic materials are repelled by magnets. **[1]**
Reason (R): Magnetic materials are attracted towards a magnet, whereas non-magnetic materials are not attracted.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

20. **Assertion (A):** The process of making magnet from iron is natural method. **[1]**
Reason (R): There are different shapes of magnet like bar magnet, horseshoe magnet, ball or cylindrical ended magnet.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

21. **Assertion (A):** Substances having the property of attracting pieces of iron or some other magnetic substances are called magnets. **[1]**
Reason (R): Magnetite is called natural magnet.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

22. **Assertion (A):** Materials that are not attracted towards magnet are called nonmagnetic. **[1]**
Reason (R): Non-magnetic materials cannot be converted into magnets.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

23. **Assertion (A):** A freely suspended magnet always points in the east-west direction. **[1]**
Reason (R): Like magnetic poles repel each other, unlike magnetic poles attract each other.

a) Both A and R are true and R is the correct explanation of A.
b) Both A and R are true but R is not the correct explanation of A.
c) A is true but R is false.
d) A is false but R is true.

24. **Assertion (A):** A magnet has two poles, the North pole and the South pole, that can exist independently. **[1]**

Reason (R): Magnetic poles always come in pairs, and a single North pole or a single South pole cannot exist.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

25. Identify the recommended storage practice for maintaining the strength of a magnet. **[1]**

26. If a magnet is freely suspended, what alignment will it take? **[1]**

27. How many poles does a bar magnet have? What are their names? **[1]**

28. How would heating a magnet affect its properties? **[1]**

29. How can you determine north and south poles of a bar magnet? **[1]**

30. Identify the difference between magnetic and non-magnetic materials with an example for each. **[1]**

31. A bar magnet has no markings to indicate its poles. How would you find out near which end its North pole is located without using another magnet? **[1]**

Section B

32. Match the following : **[2]**

Column A (Terms)	Column B (Descriptions)
(a) Magnetic Poles	(i) Includes both North and South
(b) Magnetic Materials	(ii) Attracted towards a magnet
(c) Non-Magnetic Materials	(iii) Not attracted towards a magnet
(d) Compassing Effect	(iv) Indicates the north-south direction

33. Evaluate the role of natural and artificial magnets in historical and modern navigation. **[2]**

34. Design a simple experiment to demonstrate the repulsion and attraction of magnet poles. **[2]**

35. Analyze the evolution of magnets from natural lodestones to modern artificial variants. **[2]**

36. How might one distinguish a magnetic material from a non-magnetic one using a magnet? **[2]**

37. Design an experiment to explore the interactions between a compass needle and a bar magnet, detailing the expected outcomes. **[2]**

38. How can the alignment of a freely suspended magnet help in navigation? **[2]**

39. Critically analyze the reasons why magnets should not be stored near electronic devices like mobile phones. **[2]**

Section C

40. Explain the importance of proper storage for magnets and outline the steps involved in storing them correctly. **[3]**

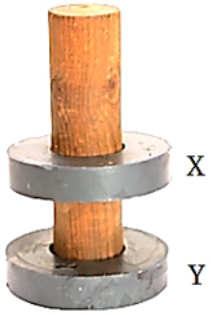
41. Describe the concept of artificial magnets and their significance in everyday objects. **[3]**

42. Evaluate the reasons why heating or dropping a magnet negatively affects its magnetic properties. **[3]**

43. How could the polarity principle of magnets be applied in crafting practical devices or tools? **[3]**

44. Imagine inventing a new type of compass using the principles of magnetism. Describe your design and its unique features. **[3]**

45. Two ring magnets X and Y are arranged as shown in Fig. It is observed that the magnet X does not move down further. What could be the possible reason? Suggest a way to bring the magnet X in contact with magnet Y, without pushing either of the magnets. **[3]**



Two ring magnets

46. Explain how magnetic compasses have evolved from the use of naturally occurring materials to modern-day variants. **[3]**

Section D

47. **Read the following text carefully and answer the questions that follow:** **[4]**

The materials which are attracted towards a magnet are called magnetic materials. The metal iron is a magnetic material. Nickel and cobalt are other metals that are also magnetic. Some of their combinations with other metals are also attracted towards magnets. The materials which are not attracted towards a magnet are called non-magnetic materials.

a. Name one metal that is a magnetic material. (1)

b. What are materials that are not attracted to a magnet called? (1)

c. List three metals that are magnetic materials. Why are they called magnetic materials? (2)

OR

What is the difference between magnetic and non-magnetic materials? Give one example of each. (2)

48. **Read the following text carefully and answer the questions that follow:** **[4]**

Air exerts pressure in all directions. This can be observed in our daily life. When we fill air into a balloon, it expands because the air inside pushes outward. When air is removed from a plastic bottle, the bottle gets crushed as the air pressure outside becomes greater. The pressure exerted by the air around us is called **atmospheric pressure**. It is caused by the weight of the air above us acting on everything on Earth.

Questions:

a. What is the pressure exerted by the air around us called? **(1)**

b. Why does a balloon expand when air is filled into it? **(1)**

c. Explain with an example how air pressure acts in daily life. **(2)**

OR

Describe what happens when air is removed from a plastic bottle and explain why. **(2)**

49. **Read the following text carefully and answer the questions that follow:** **[4]**

A freely suspended magnet comes to rest along the **north-south direction**. The end of the magnet that points towards the north direction is called the **North-seeking pole** or the **North pole** of the magnet. The other end that points towards the south direction is called the **South-seeking pole** or the **South pole** of the magnet.

A freely suspended magnet rests along the north-south direction because **Earth itself behaves like a giant magnet**. If we repeat this activity using a small **iron bar**, we observe that it can rest in any direction, not necessarily north-south. This means that only magnets have this property.

This property helps us **test whether a piece of metal is magnetic** and is also used to **find directions**. Based on this principle, a device called a **magnetic compass** was developed. It has a magnet in the shape of a **needle that can rotate freely**. The needle always points in the **north-south direction**, helping in navigation.

Questions:

a. What do we call the poles of a freely suspended magnet that come to rest along the north-south direction? **(1)**

b. Why does a magnet rest in the north-south direction? **(1)**

c. Explain how a compass is used to find directions at a new location. **(2)**

OR

How can we test whether a piece of metal is a magnet or not? **(2)**

Section E

50. Differentiate magnetic and non-magnetic materials. Evaluate how a magnet can help classify materials using suitable examples. **[5]**

51. Critique the hypothesis that breaking a magnet can result in two separate magnets each having only one pole. Use evidence from the properties of magnets to support your reasoning. **[5]**

52. What prompted scientists to transition from using naturally occurring magnets to creating artificial magnets? Discuss the benefits associated with this transition. **[5]**

53. How does the interaction between two magnets work? Conduct an experiment. **[5]**

54. Why does a magnet always have two poles? What happens when a magnet is broken? **[5]**

55. A mechanic keeps dropping tiny screws while repairing gadgets. How can he solve this problem using magnets? **[5]**

56. Describe the steps involved in using a magnetic compass to determine direction and discuss how this process is reliable even in modern navigation. **[5]**

Section F

57. **State whether the given statement is True or False:** **[9]**

(a) The North pole of some magnets is marked with a white dot. **[1]**

(b) A wooden material can be magnetised. **[1]**

(c) Poles of a bar magnet are located in the middle. **[1]**

(d) Lodestone is a natural magnet. **[1]**

(e) A magnet can exist with only a single North pole or South pole. **[1]**

(f) When the South pole of the magnet is near the North pole of the compass needle, the needle moves away. **[1]**

(g) The two ends of a bar magnet are known as the equator. **[1]**

(h) Materials like iron, nickel, and cobalt are classified as non-magnetic materials. **[1]**

(i) Opposite poles of two magnets repel each other whereas similar poles attract one another. **[1]**

Section G

58. **Fill in the blanks:** **[10]**

(a) The compass needle is also a _____. **[1]**

(b) A magnet has _____ poles. **[1]**

(c) Artificial magnets have _____ shapes. **[1]**

(d) Crane picks up iron from the junk food uses _____. **[1]**

(e) The substances having the property of attracting iron are known as _____. **[1]**

(f) The attraction of a magnet is strongest at its _____. **[1]**

(g) Stickers, pin holders and pencil boxes have _____ fixed inside. **[1]**

(h) Crane that picks up iron from the junk yard uses _____. **[1]**

(i) Magnetic materials are the materials that are _____ towards a magnet. **[1]**

(j) In some magnets, the North and South poles are marked as _____. **[1]**